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Career planning and entrepreneurial intention guidance for college students based on fuzzy decision support system

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Abstract

Due to the inclusion of subjective factors and ambiguity, the decision-making processes that are currently in use are useless. Increasing the number of resources that college students have access to in order to properly begin their professions and start their own businesses is the objective of this study. A flexible decision-support system that makes use of fuzzy logic to bring together a variety of different items is referred to as the FuzzyLogic Career and Entrepreneurship Guidance Framework (FLC-EGF). One's decision may be influenced by a variety of factors, including their level of comfort with taking risks, their hobbies and abilities, as well as the current situation of the labor market. Combining qualitative and quantitative methods of data analysis results in the generation of personalized concepts for each and every student. FLC-EGF seems to fulfill people's requirements ninety percent of the time, according to user testing, and it has the potential to increase estimates of company success by fifteen percent. According to the results of this research, the framework is beneficial in supporting students in making well-informed judgments about their future careers and in actively pursuing business opportunities. As a means of bridging the gap between what students learn in school and what businesses need more of, FLC-EGF provides assistance to outstanding workers and successful company owners. We give financial support to groups that assist people in determining their life's purpose, provide encouragement to those who are interested in beginning their own enterprises, and invest money in the labor force in order to achieve this objective.

Keywords Fuzzy logic, Decision support system, Personalized guidance, Entrepreneurship education, Workforce development, Risk tolerance

1 Introduction

To do well in today's difficult employment market, college students need help in planning their careers and starting their own businesses. Students should think about their entrepreneurial spirit and the work options that are accessible to them as the global labor market is always changing [1]. It may not be adequate to have broad guiding concepts since it may be challenging to predict professional and business decisions, as well as personal likes and expectations. We need new ways to help each student since our

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present ones aren't working and we can't deal with all the disparities between students [2]. People may choose careers they would love by organizing their thoughts about their interests, skills, and goals. This is what career planning does. Students may be sure that this method [3] will provide them a clear and easy way to go to their future careers.

People who wish to create and run successful businesses may benefit from entrepreneurial goal coaching, which may help them have the proper mindset, knowledge, and skills [4]. To get students ready for regular jobs and jobs as entrepreneurs, these problems need to be addressed [5]. Personalized, flexible guiding systems are very important in today's world since technology is continuously developing and industries are always changing [6]. Static models are less useful when they don't take into account how people are different and how the market changes over time [7]. Unfortunately, there aren't enough personalized and appropriate ways to provide business and professional advise. People frequently end up with broad strokes that don't take their specific profiles into account, don't manage uncertainty when making judgments, can't adjust to new situations, or don't bring together diverse aspects [8]. This disparity shows how important it is for a system to be able to deal with these problems and change the aid it delivers to students when it needs to.

Most of the time, research and guidance systems use stringent, rule-based methods that don't take into consideration that people are different and that things may change [9]. Students are frequently forced to follow specific professional paths without thinking about what they want to do or the dangers that come with being an entrepreneur [10]. Fuzzy logic isn't used by most modern navigation systems. Instead, they employ models that can't be changed to fit the demands of various businesses or the aspirations of individual pupils [11]. Please fill out the portions so we can make our career and business coaching better.

The purpose of this research is to use fuzzy logic as a technology [12] to solve the issue of personalized student help. Fuzzy logic may include subjective data, make complicated relationships clearer, and provide useful solutions in real-life situations where choices need to be made [13]. This technique fills in the holes left by previous answers, making it a robust, adaptable response that may be used in many different scenarios. The career and entrepreneurship guidance framework (FLC-EGF) employs fuzzy decision-support systems [14] to assist individuals in making choices in complex settings. Students may use this framework to help them discover a job that fits their skills, interests, and career objectives better [15]. The main findings of this study are:

- The FuzzyLogic Career and Entrepreneurship Guidance Platform (FLC-EGF) combines the skills, hobbies, and market trends to find out how much risk one can take and what the best career path is for you. As an expert, it's hard to make decisions, but our all-encompassing method makes it easier.
- People may make more objective decisions about their professions and fewer subjective ones if they used fuzzy logic in the FLC-EGF. The framework may change its ideas to fit the needs of each learner, which is not achievable with rigid choice models.
- FLC-EGF's selections are more accurate and dependable when they use the Mamdani Fuzzy Inference System, especially when the data isn't particularly clear. It's simpler to predict how well a firm will do when concepts and user goals are more closely linked than when they are more conventional.

- Making an FLC-EGF that may be used for job searches, career counseling, and entrepreneurial programs. This resource gives students real-world information that will help them do well in their careers, motivate them to establish their own enterprises, and link what they learn in school with what businesses need.

This is the structure of the paper: Sect. 2 speaks on the challenges with the present ways of planning a career and aiding entrepreneurs. In Sect. 3, we talk about the FLC-EGF and all that goes into it. In Sect. 4, we talk about the experiments that show how the framework may assist make business proposals and predictions of success more accurate. After going over the basics in Sect. 5, we talk about what will happen next, such how schools could use advanced fuzzy algorithms.

2 Related works

Using public opinion networks as a case study, Yue [16] conducted research to analyze the viewpoints of college and university students about their job and future ambitions. A description is given of the fuzzy evaluation technique, which includes a discussion of its benefits and drawbacks for decision-making, as well as an explanation of how it may be used to learn what people believe about networks. The purpose of this research is to evaluate the relationship between students' entrepreneurial education and their desired outcomes, activities, and the development of new businesses. Within the scope of this project, we are working on fuzzy assessment in order to simplify and improve the reliability of the process of gathering public input on university networks. The purpose of this model is to give researchers and practitioners with a comprehensive and methodical evaluation approach that can be used to improve education and the growth of students.

The purpose of the twenty-item scale developed by Viswanath and Kumar [17] was to ascertain the influence that perceived obstacles have on the choices that social entrepreneurs make on their careers. The item analysis, exploratory factor analysis (EFA), and confirmatory factor analysis (CFA) methods were used in the construction of this quantitative scale. This paper investigates the aspects of the examination that aspiring social entrepreneurs utilize to guide their career choices prior to being accepted into the entrepreneurship program. Having the ability to perform something, having the appropriate knowledge and tools, being valuable, and having a standard meaning are the four components that make up these aspects. Because it contributes to the body of literature on entrepreneurial cognition, simplifies the process of conducting quantitative research and testing models, and assists with the development and evaluation of educational programs, the social entrepreneurship scale is an extremely useful instrument for academics, educators, and policymakers who are working in the field of social entrepreneurship.

Within the framework of their proposal for the FC-JS2-TSC approach, Xiao and Liu [18] integrated Takagi–Sugeno fuzzy inference with Cuckoo search optimization. Personalized advice and an integrated job search system are the goals—the FC models job indecision and market fluctuations with fuzzy logic and sets. Methods are tailored to student profiles and preferences. Experimental validation reveals that FC-JS2-TSC outperforms previous job strategy selection and results approaches. The program streamlines job searches by enabling data-driven, personalized decision-making. Integrated optimization and fuzzy logic's uncertainty handling give students the finest job search odds.

Li and Xiao[19] used mass innovation and entrepreneurship to develop innovative and entrepreneurial students in colleges and universities. They offer thorough training for college students to foster innovation and entrepreneurship. Classical multiple-attribute group decision-making (MAGDM) is used to assess these kids' originality and entrepreneurship. Based on probabilistic uncertain linguistic sets (PULTSs) and the combined compromise solution approach, the PUL-CoCoSo method is suggested for MAGDM. PULTSs extract attribute weights, and an application evaluating college students' creativity and entrepreneurial abilities uses these algorithms.

Jin [20] reported that personalized training is essential for addressing individual differences in educational attainment, perspectives, employment choices, and confidence in college enrollment. This study presents an emperor penguin search-assisted artificial neural network (EPS-ANN) method for predicting college students' career choices. Training employs a dataset of student academic transcripts, preprocessed for feature evaluation, using linear discriminant analysis (LDA) to extract relevant features. Modifying the EPS enhances the ANN parameters. It has been tested on Python and compared to various ways of doing things. The EPS-ANN method is better than its predecessors at predicting what college students will do for a living because it stresses the necessity of using a variety of data and getting hands-on experience.

Sisu et al. [21] investigated students' inclination towards entrepreneurship with a combination of fuzzy-set qualitative comparative analysis and partial least squares structural equation modeling. The findings indicate that several initiatives may inspire students to create their own businesses, such as mentoring programs, innovation labs, entrepreneurship lectures, and business incubation initiatives. This research has considerable implications for instructional practices and policy about the future of entrepreneurial education. The author posits that unique educational settings, such decision theaters and entrepreneurial boot camps, are essential for fostering creativity and encouraging risk-taking. This research underscores the need of using several techniques in entrepreneurship education.

One way to tackle social and environmental problems is to encourage long-term economic development. One approach to achieve this is to get college students to create enterprises that are good for the environment [22]. Much research has examined self-efficacy, creativity, education, and financial assistance, but it typically ignores the complex interplay among these factors. Based on 207 questionnaires from Chinese coastal colleges, entrepreneurial inventiveness, green cognition, financial support, future self-continuity, entrepreneurship culture, and education were the main pathways to high green entrepreneurial intention. By teaching innovative thinking and green entrepreneurship, college instructors can use the findings to boost students' entrepreneurial creativity and green cognitive capacity.

Zhou [23] reported that Chinese universities' Double First Class construction model cultivates top talent. Local universities are adjusting by upgrading curricula, investing in practical professional instruction, and encouraging innovation and entrepreneurship. MADM is used to evaluate local college students' "Entrepreneurship and Entrepreneurship" career skills. MADM is addressed with CoCoSo and information entropy. The test use triangular fuzzy neutrosophic sets (TFNSs) to express uncertainty. In this study, we provide the numerical results and use decision comparisons to verify the TFNN-CoCoSo approach.

Chaw et al. [24] investigated immigrant entrepreneurs, pinpointing essential factors for assessment, including social support, cultural competence, self-efficacy, favorable personality traits, and a conviction in the validity of hierarchy. This research assesses immigrant families from four nations using structural equation modeling (SEM) and multiple-criteria decision-making (MCDM). The findings suggest that immigrant entrepreneurs might improve their economic success by honing certain skills and traits. When developing strategies to aid businesses owned by immigrants, managers may want to keep what the study discovered in mind. The study is especially relevant on a global scale due of its distinctiveness.

Yang [25] changed how students and schools interacted with each other to find out what traits made it more likely for students to start their own businesses. They also examined the factors that lead to these occurrences. The method helps local governments, colleges, and college students reach their business goals by employing the analytical technique's judgment matrices for each layer to find important aspects, provide theoretical guidance, and give tools for making models. We utilized analytical hierarchy processing (AHP) to look at how to build an index for assessing innovation and entrepreneurship education that included a "four-in-one" survey, expert consultation, and hands-on learning. The results are in Table 1.

An investigation that was conducted not too long ago suggests that frameworks for career and entrepreneurial decision-making should include fuzzy logic and hybrid intelligence systems. By using genetic algorithms, fuzzy logic, neural networks, and machine learning, Mitrakas et al. [26] showed the management of nonlinearity, uncertainty, and dynamic interactions in occupational risk assessment. This was accomplished by the use of these techniques. The findings of their study demonstrate that mixed fuzzy logic models result in calculations that are both more accurate and simpler to comprehend. For the purpose of determining the extent to which persons working in the technology industry experience joy, Sultan et al. [27] used a complete fuzzy language regression model. The findings of this study demonstrate that it is possible for multi-criteria fuzzy decision systems to effectively capture human subjectivity, in contrast to typical deterministic regression models. In order to generate recommendations that are both clear and concise, the FLC-EGF framework makes use of Mamdani fuzzy inference. This might be of assistance to those who are at a loss as to what they should do with their businesses or professions. The FLC-EGF algorithm is superior to the more difficult hybrid algorithms in terms of accuracy, ease of comprehension, and ease of application. Additionally, it is simpler to modify and has lower operating expenses. It is an excellent option for regions where individuals may go to school and get assistance with their work.

Recent developments in fuzzy-set theory have led to sophisticated measures of distance and divergence that enhance the robustness, discrimination, and uncertainty modeling of decision systems. These concepts are very helpful for customized career and business coaching. Zhu et al. [28] introduced α-divergence measures for image and interval-valued picture fuzzy sets, which offer flexible sensitivity modification and a superior assessment of alternatives under complicated ambiguity. Liu et al. [29] created distance measures for complicated Fermatean fuzzy sets that may reflect deeper hesitation and nonlinear uncertainty, enabling more distinction across student profiles and enhancing the precision of rule selection and classification. Zhu et al. [30] also showed how valuable distance measures are for diagnosing and discovering trends in complicated image

Table 1 Summary of literature survey

Authors	Objective	Methods	Findings	Advantages	Disadvantages	Research gap
Yue [16]	Application of fuzzy control algorithm in career planning and entrepreneurship	Fuzzy control algorithm	Enhanced decision-making in career planning and entrepreneurship	Provides precise and flexible guidance	Limited real-time adaptability	Lack of scalability and comparison with non-fuzzy methods
Viswanath and Kumar [17]	Development of a social entrepreneurship career decision-making scale	Scale development and validation	Reliable scale to assess social entrepreneurship decisions	Focuses on social entrepreneurship	Limited applicability to general career planning	Need for cross-cultural validation
Xiao and Liu [18]	Optimization of job search strategies for college students	Fuzzy logic with cuckoo search algorithm	Improved efficiency in job search strategies	Combines fuzzy logic with metaheuristic optimization	Computationally resource-intensive	Limited diversity in testing across demographics
Li and Xiao [19]	Evaluation of entrepreneurial skills using advanced CoCoSo method	Combined compromise solution (CoCoSo) method	Improved entrepreneurial skill evaluation under uncertain conditions	Effective for multi-attribute decision-making	Requires real-world testing	Need for application to dynamic and evolving skills
Jin [20]	Prediction of career inclinations with personalized suggestions	Neural network model	High accuracy in predicting career inclinations and personalized guidance	Provides tailored career advice	High computational complexity	Lack of integration with fuzzy logic for enhanced recommendations
Sisu et al. [21]	Analysis of enablers for entrepreneurial intentions	PLS-SEM and fuzzy set qualitative comparative analysis (fsQCA)	Identified key enablers for entrepreneurial intentions	Combines structural and fuzzy approaches	Complex for non-experts	Need for longitudinal studies and broader applicability
Cai et al. [22]	Promotion of green entrepreneurial intentions among college students	Fuzzy set qualitative comparative analysis	Identified key factors influencing green entrepreneurial intentions	Focuses on sustainability	Limited to green entrepreneurship	Lack of application to other entrepreneurial domains
Zhou [23]	Evaluation of entrepreneurship career abilities with triangular neutrosophic numbers	Triangular neutrosophic numbers	Novel method for evaluating entrepreneurship capabilities	Innovative evaluation approach	Requires understanding of neutrosophic theory	Need for validation with diverse real-world datasets
Khaw et al. [24]	Influence of generational status on entrepreneurial intentions	Structural equation modeling and multicriteria decision-making	Highlights generational differences in entrepreneurial intentions	Multidimensional analysis	May lack generalizability across cultures	Further exploration of global generational dynamics
Yang [25]	Exploration of factors influencing innovation and entrepreneurship intentions	Analytic hierarchy process (AHP)	Prioritization of key factors influencing entrepreneurial intentions	Clear and structured approach	Lacks real-time adaptability	Need for integration with fuzzy logic for dynamic decision-making

fuzzy environments, which may improve FLC-EGF's capacity to manage overlapping student traits and fuzzy borders. For interval-valued image fuzzy sets, Zhu et al. [31] have presented a fuzzy TOPSIS technique that uses new divergence metrics for trustworthy risk assessment and prioritizing. Similar to FLC-EGF's assessment of career and entrepreneurial prospects, this method improves decision stability and interpretability. All things considered, these advancements demonstrate how sophisticated fuzzy distance and divergence models are becoming more and more important for developing data-driven, interpretable, and flexible decision-support systems such as FLC-EGF, especially in intricate human-centered fields.

3 FuzzyLogic career and entrepreneurship guidance framework

For college students, this FLC-EGF technology, which is based on fuzzy logic, might be of assistance in determining what they want to do once they graduate from college. In order to do this, the GUESS dataset [32] is used. Surveys have been conducted with college students from all around the world who are interested in beginning their own successful enterprises. These surveys inquire about their aspirations, personality traits, and potential careers in the future. The FLC-EGF includes Data Collection, Data Preprocessing, Fuzzification Process, Rule Application, Fuzzy Inference, Defuzzification, and Integration & Action Plan. Student data must be organized, normalized, cleaned, and converted into fuzzy sets. The Fuzzification process captures data uncertainty. The final stage combines professional and entrepreneurship insights to develop a comprehensive action plan. The FLC-EGF uses fuzzy logic to make individualized job suggestions. Starting with the START node, the process includes data collection, preprocessing, and fuzzification.

The current implementation of the FLC-EGF primarily applies the adaptation mechanism to the defuzzification process, the same approach can be expanded to other stages of a fuzzy inference system, such as the fuzzification and rule weighting stages, further to improve interpretability, flexibility, and overall decision accuracy.

Figure 1 depicts the fuzzy logic-based decision support system that is used for the purpose of assisting college students in conducting career and business planning. Initially, the system initiates the process of gathering and preparing student data, which is subsequently turned into fuzzy inputs. The system routes the data through either the career path track or the entrepreneurship track, depending on the student's profile and the preferences they have expressed. To determine whether or not anything is suitable, fuzzy rules are utilized, and then the findings are compiled and defuzzified in order to produce recommendations that may be put into action. After that, the information that was gathered from both methods is combined in order to create a comprehensive coaching package. One of the advantages of the system is that it has the potential to become more effective at generating new ideas as it gathers more information or gain more understanding about the pupils.

3.1 Data collection and preprocessing

3.1.1 Data collection

Investigators from all over the world make use of the GUESS dataset [32] to examine the entrepreneurial spirit and aspirations of college students, as well as the influence of their environment on their cognitive abilities. The GUESS dataset is a great resource



Fig. 1 Flow chart for FLC-EGF

Table 2 Core input variables for FLC-EGF

Variable code	Attribute name	Type	Description
AGE	Age	Continuous	Student's age in years
GND	Gender	Categorical	Male/Female/Other
MAJ	Major discipline	Categorical	STEM/Business/Arts/Other
EIS	Entrepreneurial interest score	Numerical	Likert-scale score (1–7) indicating interest in entrepreneurship
CSS	Career stability score	Numerical	Likert-scale score (1–7) indicating preference for job security
LS	Leadership skill	Numerical	Self-assessed leadership ability (scale 0–100)
IT	Innovation tendency	Numerical	Self-assessed creativity and innovation index (scale 0–100)
RT	Risk tolerance	Numerical	Willingness to take risks (scale 0–100)
RCS	Role compatibility score	Numerical	Perceived match between personal traits and role requirements (0–100)

for research in the fields of cross-cultural entrepreneurship and career coaching since it contains both quantitative and qualitative answers from standardized traits across more than fifty countries.

In Table 2, The FLC-EGF utilizes well-defined features from the GUESS dataset to capture college student's key qualities that affect their career or entrepreneurial paths. The

fuzzy logic inference mechanism relies on these demographic, cognitive, behavioral, and attitudinal factors. For the purpose of simplifying the process of developing customized recommendation systems and making decisions when things aren't quite obvious, we choose each component. These characteristics are indicative of the degree to which an individual is committed to founding a business or following a certain professional path. In addition, they reveal the way in which people think and the choices that they make.

3.1.2 Data preprocessing

A significant amount of work is performed on the GUESS dataset by the FLC-EGF before it is applied to the model, which results in an improvement in fuzzy logic modeling. Data normalization, cleansing, and feature engineering are used for reliability and consistency. Min–Max normalization standardizes numerical variables, removing magnitude bias. Cleaning data comprises deleting incomplete records, detecting and removing outliers using the Interquartile Range (IQR) approach, and replacing missing values with the mean and mode. Using composite indicators from existing variables, feature engineering improves FLC-EGF decision-making. These indicators represent latent traits that directly affect career and entrepreneurship. The Entrepreneurial Propensity Index (EPI) measures students' cognitive and behavioral entrepreneurial propensityCareer Stability Preference (CSP) connects students' preferences for orderly, stable, and predictable jobs with stability-oriented careers. FLC-EGF provides complete career and entrepreneurial advice. A structured translation of raw student variables from the GUESS dataset allows the FLC-EGF to give tailored suggestions under uncertainty. Use these equations for smoothing, outlier detection, normalization, and feature extraction.

$$x'_t = \frac{1}{n}\sum_{i=0}^{n-1} x_{t-i} \tag{1}$$

In Eq. 1, x'_t is represented as a smoothed sensor reading at time *t.n* is defined as a window size and x_{t-i} is represented as Original sensor readings.

$$z = \frac{x-\mu}{\sigma} \tag{2}$$

The Z-score approach subtracts outliers by comparing a data point's mean to its standard deviation in Eq. 2. The following step, normalization, uses Eq. 3 to compare all variables.

$$X_{norm} = \frac{(X - X_{min})}{(X_{max} - X_{min})} \tag{3}$$

Feature Engineering: FLC-EGF employs the Entrepreneurial Propensity Index (EPI) and Career Stability Preference (CSP) designed variables to increase inference accuracy and distinguish pupils with high entrepreneurial desire from those with stable employment.

$$EPI = 0.4 \cdot RT + 0.3 \cdot IT + 0.3 \cdot LS \tag{4}$$

$$CSP = 0.5 \cdot Security + 0.3 \cdot Predictability + 0.2 \cdot Structure \tag{5}$$

In Eqs. 4 and 5, Composite indicators are crucial in distinguishing students with high entrepreneurial intent from those who prefer stable, structured careers.

The FLC-EGF's fuzzification module uses a core feature set (F) consisting of variables such as Risk Tolerance(RT), Innovation Tendency(IT), Leadership Skill(LS), Entrepreneurial Interest Score(EIS), and Career Stability Score(CSS) to represent a student's entrepreneurial and career personality profile, as per Equation (6).

$$F = \{RT, IT, LS, EIS, CSS\} \tag{6}$$

The correlation-based approach for feature selection involves calculating the correlation between the features as shown in Eq. 7:

$$C\left(f_i, T\right) = \frac{Cov\left(f_i, T\right)}{\sigma_{f_i}.\sigma_T} \tag{7}$$

In Eq. 7, $C\left(f_i, T\right)$ is the correlation between the feature f_i and the target output is T. Features are cleaned and normalized for the fuzzy system. $D_{processed}$: Aggregating the dataset, normalizing it using $RT, IT, LS, EISandCSS$ norms, and counting emergency scenarios are the steps. After cleaning and normalizing, the system input is prepared. The fuzzy model is created from system input to anticipate system outputs. The procedure makes the system robust and efficient, minimizing errors and improving performance, as shown in Eq. 8.

$$\begin{aligned} &D_{processed} = \{(RT_{norm}, IT_{norm}, LS_{norm}, EIS_{norm}, CSS_{norm})_i \,|i = 1, 2, \ldots, N \;\; where \\ &Input = \{RT_{norm}, IT_{norm}, LS_{norm}, EIS_{norm}, CSS_{norm}\} \end{aligned} \tag{8}$$

The FLC-EGF requires preprocessed and engineered data from the GUESS dataset, shown in Fig. 2. Technical and interpersonal skills, risk tolerance, innovation, leadership, EPI, and CSP are included. The preprocessing pipeline organizes student survey data for FLC-EGF fuzzification and inference. These engineered elements improve the system's tailored guiding and nuanced decision-making.

3.2 Fuzzification process in FLC-EGF

The Mamdani Fuzzy Inference System rule assessment requires fuzzy linguistic variables for crisp input values, including interest level, skill rating, market trend alignment, and risk tolerance. Membership functions for GUESS dataset input variables are triangular and trapezoidal.

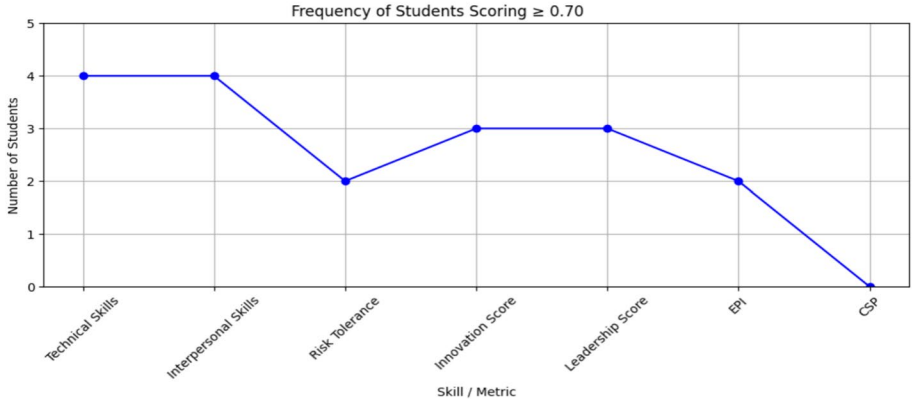


Fig. 2 Frequency of student scoring

Table 3 lists four GUESS dataset input variables: Interest_Level, Skill_Competency, Market_Trend_Alignment, and Risk_Tolerance. Each variable is fuzzified using appropriate membership functions, such as triangular for variables with exact anchor points and trapezoidal for ranges of values that may fully satisfy a fuzzy term. Using multi-valued logic representation improves the system's capacity to make individualized and adaptable career and entrepreneurial recommendations. Graphs for each input variable's fuzzy sets are offered to visualize membership functions.

3.2.1 Membership functions

In the Fuzzification Process, membership functions assign a degree of membership ($\mu \in [0, 1]$) to crisp input values, transforming them into fuzzy sets. Fuzzy Logic systems use Triangular and Trapezoidal membership functions for simplicity and computing efficiency. Let $\mu_X(x)$ is the Degree of membership of value x in fuzzy set X.

$$\mu_{Tri}(x; a, b, c) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{b-a}, & \mathrm{a} < \mathrm{x} \leq \mathrm{b} \\ \frac{c-x}{c-b}, & b < x \leq c \\ 0, & x > c \end{cases} \tag{9}$$

In Eq. 9, A Triangular Membership Function is a piecewise linear function with three parameters: $a, b, andc$. The function is piecewise linear, rising to the peak and decreasing symmetrically or asymmetrically, with a denoting the left endpoint, b the peak point, and c the right endpoint.

$$\mu_{Trap}(x; a, b, c, d) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{b-a}, & a < x \leq b \\ 1, & b < x \leq c \\ \frac{d-x}{d-c}, & c < x \leq d \\ 0, & x > c \end{cases} \tag{10}$$

In Eq. 10, A Trapezoidal Membership Function (FLC-EGF) is a fuzzy set model that consists of four parameters: $a, b, c, andd$. Membership rises from 0 to 1 between a and b, continues at 1 between $bandc$, then drops to 0 between $candd$. Broader linguistic phrases and saturated categories like advanced skill competencies are suited for the function.

Fuzzy membership functions for Skill_Competency in the Fuzzy Logic Controller for Educational Guidance Framework are shown in Fig. 3. It shows how membership degrees connect numerical skill ratings to linguistic phrases. Membership function definitions, precise membership functions, and visualization are essential. Triangular and trapezoidal membership functions range from 0 to 5 in the script. The graphic color-codes membership functions and displays how a precise skill rating value maps to membership degrees for each linguistic phrase, including labels, title, grid, and legend. In the

Table 3 Input variables from the GUESS dataset

Variable name	Description	Type	Range	Linguistic terms
Interest_Level	Self-reported career/entrepreneurial interest	Numeric	[0, 10]	Low, Medium, High
Skill_Competency	Skill self-assessment score	Numeric	[0, 10]	Novice, Intermediate, Advanced
Market_Trend_Alignment	Industry demand vs. interest alignment score	Derived	[0, 100]	Poor, Moderate, Strong
Risk_Tolerance	Risk attitude scale	Numeric	[1, 5]	Low, Moderate, High

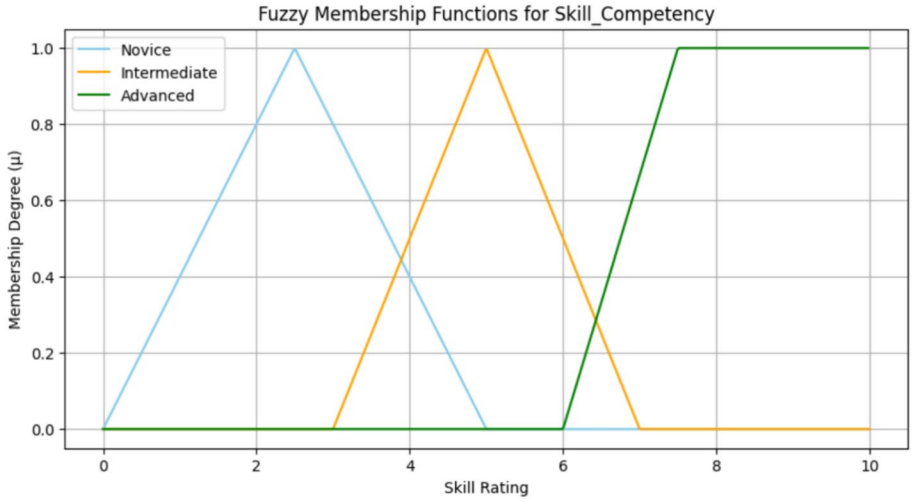


Fig. 3 Fuzzy membership function for skill competencies

Mamdani Fuzzy Inference System, rule evaluation requires the fuzzification of precise skill evaluations into fuzzy linguistic variables with varied degrees of membership.

3.3 Process path decision module

Each student's profile is dynamically assigned to a fuzzy rule set for career, entrepreneurship, or both using fuzzy logic. A basic yet effective rule-based system identifies a student's career based on interest, risk tolerance, market trend alignment, and entrepreneurship. GUESS sample dataset values show that risk-tolerant students are more entrepreneurial. Those with higher interest are more inclined to pursue occupations, and these characteristics influence the decision. Rule-based decision reasoning with clear thresholds is simple but effective.

$$Path = \begin{cases} E, & \text{if } I_e > 7 \wedge R_t > 3 \\ C, & \text{if } I_c > 7 \wedge M_a > 70 \\ B & \text{otherwise} \end{cases} \tag{11}$$

In Eq. 11, The *Path* equation represents a rule-based decision function that determines the most suitable career direction for an individual. The decision is made by evaluating a set of quantitative criteria related to entrepreneurial intent, risk tolerance, interest level, and market alignment. When the entrepreneurial intent score (I_e) is greater than 7 and the risk tolerance (R_t) exceeds 3, the model recommends the *Entrepreneurship* path. Following the market trend alignment (M_a) of 70 and the interest level (I_c) of 7 is the most effective way to go with one's professional life. In every other circumstance, it refers to the result as "Both," which indicates that it needs to have a balanced combination of professional and commercial interests. By looking into both how to behave and the things that are going on around one, this equation has the potential to present one with individualized, data-driven work suggestions.

Students may utilize the FLC-EGF and GUESS databases to determine the most efficient routes to take in order to reach their desired destination. The approach may serve as a foundation for students to gain professionalism, develop their own businesses, or just achieve a more beneficial balance between their personal lives and their working lives. Table 4 presents five different instances of student work, which are as follows: In

Table 4 Process path decision from the GUESS dataset

Student_ID	Interest_Level	Skill_Competency	Market_Trend_Alignment	Risk_Tolerance	Entrepreneurial_Intent_Score	Decision path
001	9	7	85	2	5	Career
002	6	6	65	4	8	Entrepreneurship
003	8	5	75	4	6	Both
004	4	3	30	2	3	Both
005	7	8	90	5	9	Entrepreneurship

light of the fact that they are highly focused and in accordance with the most recent developments in the professional world, Student 001 is actively seeking for work. It would be in 002's best interest to launch their own company if they are both intelligent and motivated. If one want to be successful in school and in the job market, like Student 003, one need to give some thought to all of the many business and career opportunities that are available to you. In order to make an informed decision on a major, Student 004, whose grades range from average to below average, should consider both of the available possibilities. Student 005 is attending school in order to get knowledge about business. As well as having a great deal of potential, they are aware of what people desire at the moment. Through the use of the specific abilities possessed by each student, the method offers them individualized instruction. For instance, it selects its routes based on factors that are not entirely clear.

In Fig. 4, the 2D Decision Region Map in the FuzzyLogic Career and The FC-EGF provides students with a wide range of career opportunities, the specifics of which are determined by the degree of risk they are ready to take and the degree to which they wish to launch their own company. Given that decision-making is a subjective process that is not always straightforward, the framework makes use of fuzzy logic. Students who are willing to take risks and have a genuine interest in business are the ideal candidates for the Orange student group. The blue option is for students who have a passion for a certain field and wish to start their own businesses. One can see a fantastic suggestion that may be useful in both directions in green. Figure 4 illustrates how the rule-based logic of the Mamdani Fuzzy Inference System operates with information that has more than one dimension. It provides ideas for making decisions that are constrained by rules in order to ensure that they are reasonable.

3.4 Career path rules and entrepreneurship path rules

Rules are used in the construction of the Mamdani Fuzzy Inference System. Expert knowledge and domain logic are both components of language-specific versions of the IF–THEN rule. A student may use these factors to help them choose whether they should acquire a job, start their own company, or do a combination of the two depending on their situation.

- Career path rules
  Students who are doing exceptionally well in school or who possess unique qualities, who have an interest in certain disciplines, and who are in step with the most current advances in the industry are given the opportunity to have more employment opportunities.

  For example:

IF	Interest_Level	IS	*High*
AND	Skill_Competency	IS	*Advanced*
AND	Market_Trend_Alignment	IS	*Strong*
THEN	Career_Compatibility	IS	*Very_High*

- Entrepreneurship Path Rules
  Students that exhibit entrepreneurial spirit, a willingness to take risks, and a capacity to generate novel concepts with limited means are the intended recipients of these regulations.

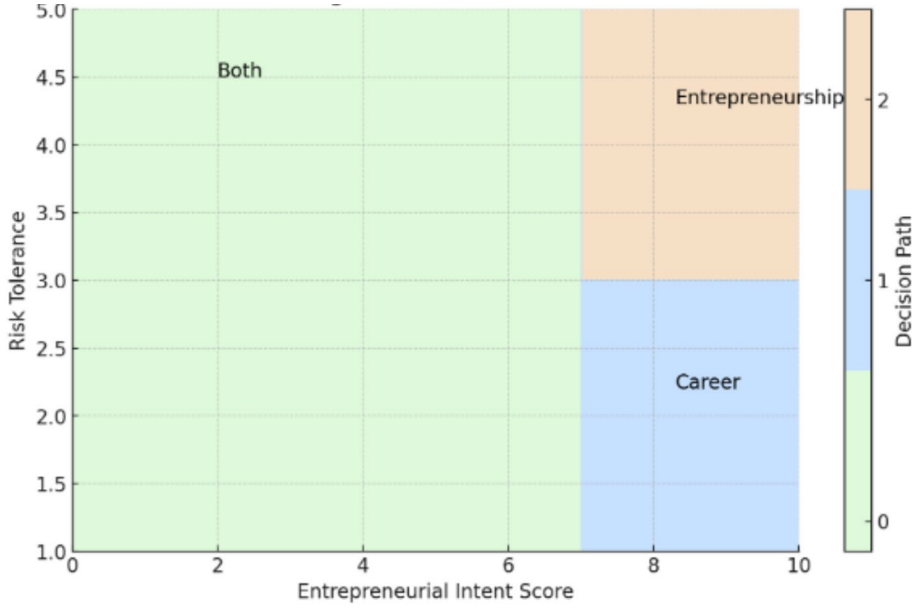


Fig. 4 2D Decision region map for career and entrepreneurship path selection in FLC-EGF

For example:

IF	Risk_Tolerance	IS	*High*
AND	Innovation_Score	IS	*High*
AND	Resource_Access	IS	*Moderate*
THEN	Business_Viability	IS	*High*

Table 5 shows how career and entrepreneurial markers blur input variables in triangular/trapezoidal membership functions. Student career or entrepreneurial fit depends on output path. Linguistic fuzzy output is defuzzed next. For precise guidance values, the Mamdani Inference Engine assesses all rules simultaneously and defuzzes final scores using Centroid.

3.5 Inference mechanisms

Mamdani Fuzzy Inference System (FIS) of FLC-EGF uses Inference Mechanism. All activated fuzzy rules are evaluated and combined into a fuzzy outcome. Fuzzy logic, inputs, and antecedent and consequent membership functions are considered. Table 6 shows that a student's interests, talents, market trends, and risk-taking aptitude must be modeled to determine employment appropriateness or entrepreneurial viability.

The mathematical representation of the final form is

$$\mu_O(y) = \max\left(\min\left(\mu_{R_1}, \mu_{C_1}(y)\right), \min\left(\mu_{R_2}, \mu_{C_2}(y)\right), \ldots\right) \tag{12}$$

In Eq. 12, for clarity and brevity, the membership functions in the fuzzy inference equation are denoted symbolically as μ_O, μ_{R_i}, and μ_{C_i}, representing the output, rule activation, and consequent membership functions, respectively. The algorithm evaluates each fuzzy rule, assesses its strength using minimum input variable degrees of membership, and truncates the fuzzy output set. The final output distribution is the maximum truncated outputs from all rules. Inference uses many concepts to give complicated, data-driven expert recommendations.

Table 5 Fuzzy rule base table

Rule No	Interest_Level	Skill_Competency	Market_Trend_Alignment	Risk_Tolerance	Innovation_Score	Resource_Access	Output path	Output value
R1	High	Advanced	Strong	Low	Low	Low	Career	Career_Compatibility = Very_High
R2	Medium	Intermediate	Moderate	Moderate	Medium	Moderate	Both	Career_Compatibility = Medium, Business_Viability = Moderate
R3	Low	Novice	Poor	High	High	High	Entrepreneurship	Business_Viability = High
R4	High	Advanced	Strong	High	High	Moderate	Both	Career_Compatibility = Very_High, Business_Viability = High
R5	Medium	Intermediate	Strong	Low	Low	Moderate	Career	Career_Compatibility = High
R6	High	Novice	Moderate	High	High	High	Entrepreneurship	Business_Viability = High
R7	Low	Intermediate	Poor	Low	Low	Low	None	Career_Compatibility = Low, Business_Viability = Poor
R8	High	Advanced	Moderate	Moderate	Medium	Moderate	Both	Career_Compatibility = High, Business_Viability = Moderate
R9	Medium	Intermediate	Moderate	High	High	High	Entrepreneurship	Business_Viability = Excellent
R10	High	Intermediate	Strong	Moderate	Medium	Moderate	Career	Career_Compatibility = High

3.6 Defuzzification process

A fuzzy inference system's ultimate step is defuzzification, which turns fuzzy output sets into a single value for actionable judgments. The Centroid approach and the Weighted Centroid method are two types of solutions that may be used to eliminate ambiguity in work trails. There are two methods that may assist one in locating the center of a fuzzy collection: the Centroid Method and the Weighted Centroid Method. The Weighted Centroid Method places a stronger emphasis on characteristics that are crucial to the success of entrepreneurs, such as the ability to be innovative and to pursue investment opportunities. As one make decisions, the most essential thing to do is to prioritize the outcomes that are most significant to the long-term objectives of the organization. This is the most critical thing one can do. For users who are interested in comprehending and working with fuzzy output sets, either approach might prove to be beneficial.

- Career path defuzzification
  The Center of Gravity Method, which is also widely referred to as the Centroid Method, is an effective method that may be used to ascertain the precise center of a fuzzy group.

$$Crisp_Career_Output = \frac{\int y \cdot \mu\left(y\right) dy}{\int \mu\left(y\right) dy} \tag{13}$$

  The integral is a method that calculates the average of all the talks that are currently available, taking into consideration the relative significance of each individual presentation. The amount of the output variable y that is a part of the group is indicated by the symbol $\mu(y)$ in Eq. 13.
- Entrepreneurship path defuzzification
  The Weighted Centroid Method is a statistical technique that may be used to determine crucial aspects of a business environment, such as the degree of importance that an idea carries and the ease with which one might collect money. It makes use of weights of importance.

$$Crisp_Entrepreneurship_Output = \frac{\int y \cdot \mu\left(y\right) \cdot W\left(y\right) dy}{\int \mu\left(y\right) \cdot W\left(y\right) dy} \tag{14}$$

  $W(y)$ is described by Eq. 14, which makes use of the Weight function. With the help of this function, various parts of the output domain are assigned the appropriate weights. This allows prioritization of outputs aligned with strategic entrepreneurship objectives.

Table 6 Max–Min composition inference mechanism

Step	Description	Mathematical expression
1. Rule evaluation	Evaluate each fuzzy rule by taking the minimum of the membership values (DoM) of the antecedents	$\mu_{rule_i} = min(\mu_{A_1}(x_1), \mu_{A_2}(x_2), \ldots, \mu_{A_n}(x_n))$
2. Implication	Truncate the output membership function (consequent) using the rule's firing strength from Step 1	$\mu_{consequence_i}(y) = min\left(\mu_{rule_i}, \mu_{B_i}(y)\right)$
3. Aggregation	Combine all individual rule outputs using the maximum operator to produce a unified fuzzy outputbreak	$\mu_{output}(y) = max_i\left(\mu_{consequence_i}(y)\right)$

Pseudocode 1 Enhanced Defuzzification for Career & Entrepreneurship Paths

```
Input:
   Y = [y₁, y₂, ..., yₙ]       # Discrete output domain values
   μ(y) = membership degree for each y
   W(y) = importance weight (applied only in entrepreneurship path)
Output:
   crisp_career_output         # Final crisp value for career path
   crisp_entrepreneur_output   # Final crisp value for entrepreneurship path
# --- Initialization ---
1. Initialize numerator_career = 0
2. Initialize denominator_career = 0
3. Initialize numerator_entrepreneur = 0
4. Initialize denominator_entrepreneur = 0
# --- Centroid Defuzzification for Career Path ---
5. For each y in Y:
6.     numerator_career ← numerator_career + (y * μ(y))
7.     denominator_career ← denominator_career + μ(y)
8. crisp_career_output ← numerator_career / denominator_career
# --- Weighted Centroid Defuzzification for Entrepreneurship Path ---
9. For each y in Y:
10.    numerator_entrepreneur ← numerator_entrepreneur + (y * μ(y) * W(y))
11.    denominator_entrepreneur ← denominator_entrepreneur + (μ(y) * W(y))
12. crisp_entrepreneur_output ← numerator_entrepreneur / denominator_entrepreneur
# --- Return Outputs ---
13. Return crisp_career_output, crisp_entrepreneur_output
```

Career and entrepreneurship path defuzzification is described in pseudocode 1. The weighted centroid-based (Entrepreneurship) and centroid-based (Career) defuzzification procedures are easily distinguished thanks to the enhanced pseudocode. Every computing step is mathematically and logically compatible with Eqs. (13) and (14), and all relevant variables are specified. By clearly demonstrating normalization and weighing procedures, the improved version increases implementation accuracy and interpretability. Changing unclear output values to sharp ones facilitates decision-making for entrepreneurial advice and career planning. The inputs are a variety of output values, fuzzy membership values, and importance or weights. The processes are initializing the output range, calculating membership values and weights, separating the Career Path results, and adding Entrepreneurship Path weighting. Weighted, the sharp output indicates the optimal career fit and entrepreneurial viability. This method turns vague choices into clear, understandable, practical recommendations. Weighted, the last Number indicates the optimal job fit and entrepreneurial possibility.

3.7 Integration and action plan module in the FLC-EGF

FLC-EGF relies on the Integration and Action Plan module to convert career and entrepreneurial projections into student action. Weighted integration assigns relevant criteria based on student profile and decision path. The module offers 3–6-month action items, 1–5-year strategic planning, resource allocation recommendations, and a skill development roadmap. The module provides proportionate guidance, time-based recommendations, resource optimization, and adaptive guidance to overcome guidance constraints. Once all is said and done, each student is given a plan that will help them put what they have learned in the classroom into practice in the real world. This plan may include obtaining a job, starting their own business, or doing both.

Figure 5 illustrates the Fuzzy Logic-Based Career and Entrepreneurship Guidance System that is now available. It is a massive plan that integrates two different ways of judgment into one. The application is able to determine, via the use of fuzzy inference and weighted integration, whether an individual would be better off pursuing a conventional career path or starting their own firm. The integrated score has a significant influence on a wide range of aspects, including the manner in which resources are employed, education and training, as well as events that are occurring now and will occur in the future. One of the last components of the concepts is a strategy that was developed via conversations with others, both verbally and numerically.

4 Results and discussion

4.1 Data study

The GUESSS is a large-scale research project that investigates the level of enthusiasm and optimism that college students from all around the globe have toward the possibility of beginning their own enterprises [32]. Hundreds of schools from fifty various countries were able to participate in the research thanks to the assistance of the Swiss School of Management Fribourg, which was present throughout the process. Through the use of a longitudinal methodology, which allows for the identification of trends, the research polled more than one hundred thousand young people in each wave. The following are some of the major human characteristics that were taken into consideration: having goals for starting a business and selecting a career, being ready to take risks, having faith in

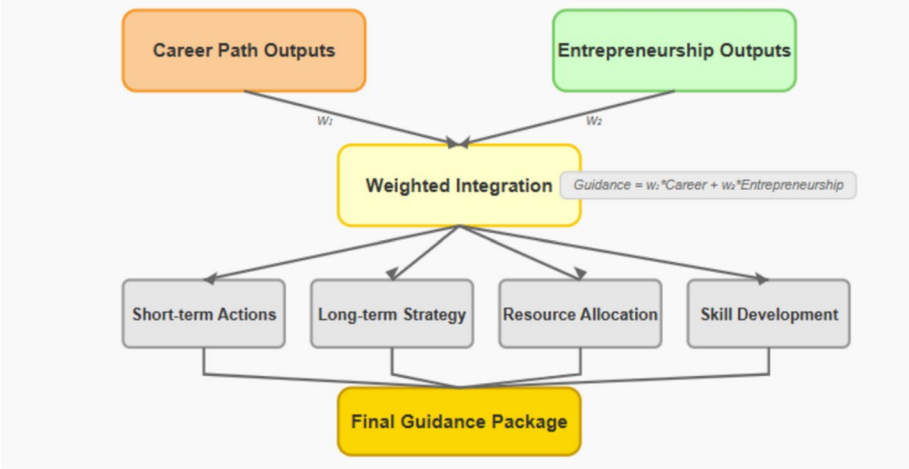


Fig. 5 Integration and action plan in FLC-EGF

oneself, and having self-confidence. The study also considers the educational and social contexts, recent entrepreneurial activities, active entrepreneurs' business performance indicators, growth aspirations, innovation orientation, and management practices.

4.2 Comparison study

College students receive updated career and entrepreneurship coaching from the Fuzzy Logic Career-Entrepreneurship coaching Framework (FLC-EGF). A Mamdani Fuzzy Inference System (FIS) with domain-specific weightings recommends customized pathways. FLC-EGF outperforms Fuzzy Logic with Cuckoo Search (FL-CS) [18], Combined Compromise Solution (CoCoSo) [19], and Triangular Neutrosophic Numbers-Based Evaluation (TNN-Eval) [23] in interpretability, linguistic reasoning, and adaptability. FLC-EGF generates user-intuitive suggestions based on personality, attitude, and context using fuzzy rule-based reasoning with weighted defuzzification. Adjustable weight integration, real-time inference, and decreased computational load are also available. FLC-EGF provides students with clearer, more explainable, and more context-sensitive career and entrepreneurial coaching than other methods. Culturally aware variables, interpretable fuzzy rules, and adjustable weight adaptation make the valuable framework for educational decision-support systems.

By comparing the performance of FLC-EGF with that of traditional algorithms (FL-CS, CoCoS, and TNN-Eval) across important criteria, Fig. 6 provides a visual representation of the comparison. Specifically, it underlines the higher specificity, MCC, solution quality, and acceptable inference time that FLC-EGF possesses. The graphic illustrates how FLC-EGF offers career and entrepreneurial counsel that is more precise, interpretable, and adaptable than traditional methods.

4.3 Specificity (%)

The dataset includes multiple student groups and various algorithms, allowing for the calculation of specificity per student group and algorithm. The equation for specificity can be extended to compute per student group and algorithm, with the Eq. 15:

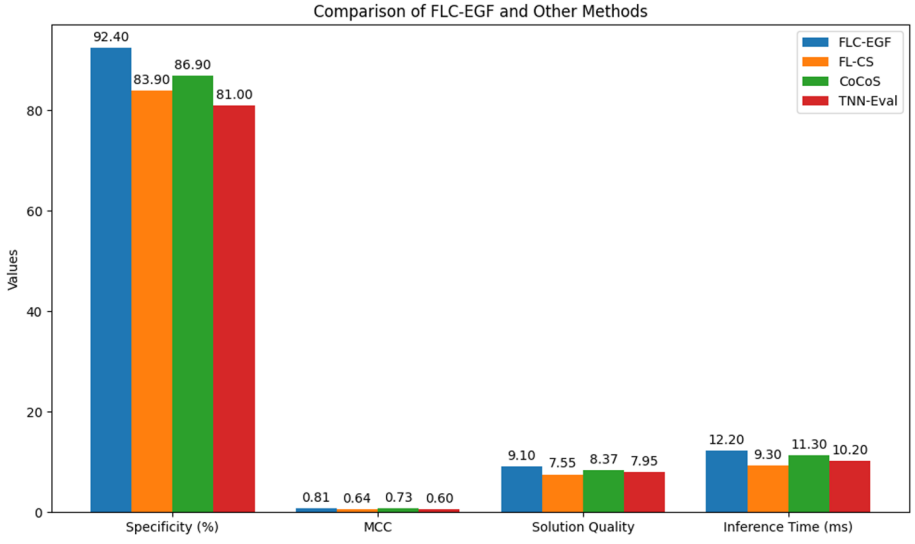


Fig. 6 Comparison of FLC-EGF and other methods

$$Specificity_{g,a} = \frac{TN_{g,a}}{TN_{g,a} + FP_{g,a}} \times 100 \tag{15}$$

In Eq. 15 , g represents the student group (e.g., L1, L2, ..., L10), a represents the algorithm (FLC-EGF, FL-CS, CoCoS, TNN-Eval). $TN_{g,a}$ is for each and every group of pupils and $FP_{g,a}$ is an algorithm. True Negatives ($TN_{g,a}$) and False Positives ($FP_{g,a}$) are both components of variable classes and algorithms (g,a), which include varying quantities of each.

In Fig. 7, the specificity percentages of four algorithms (FLC-EGF, TNN-Eval, CoCoS, and FL-CS) are shown in accordance with the career planning and entrepreneurial aims of 10 different student groups. The FLC-EGF algorithm consistently operates at a high level of specificity, with values ranging from 91.0 to 94.6%. We experimented with a variety of methods, and the FLC-EGF algorithm proved to be the most effective. Between 91.0 and 94.6% was the range of findings that were obtained for its specificity. Given that its specificity value was 94.6%, the FLC-EGF algorithm that was part of Group L6 performed much better than the other algorithms. When it comes to providing guidance on careers and businesses, the FLC-EGF framework is superior to its rivals in terms of reducing the number of false positives that it generates. It is estimated that the most effective algorithms perform around 8 to 9 percent better than the least effective ones.

4.4 MCC (Matthews correlation coefficient)

The MCC for each student group and algorithm can be extended as Eq. 16:

$$MCC_{g,a} = \frac{(TP_{g,a} \cdot TN_{g,a}) - (FP_{g,a} \cdot FN_{g,a})}{\sqrt{(TP_{g,a} + FP_{g,a})\,(TP_{g,a} + FN_{g,a})\,(TN_{g,a} + FP_{g,a})\,(TN_{g,a} + FN_{g,a})}} \tag{16}$$

In Eq. 16, $TP_{g,a}$ is the True Negative for student group g and algorithm a. $FN_{g,a}$ is the False Positive for student group g and algorithm a. $FN_{g,a}$ is the False Negative for student group g and algorithm a.

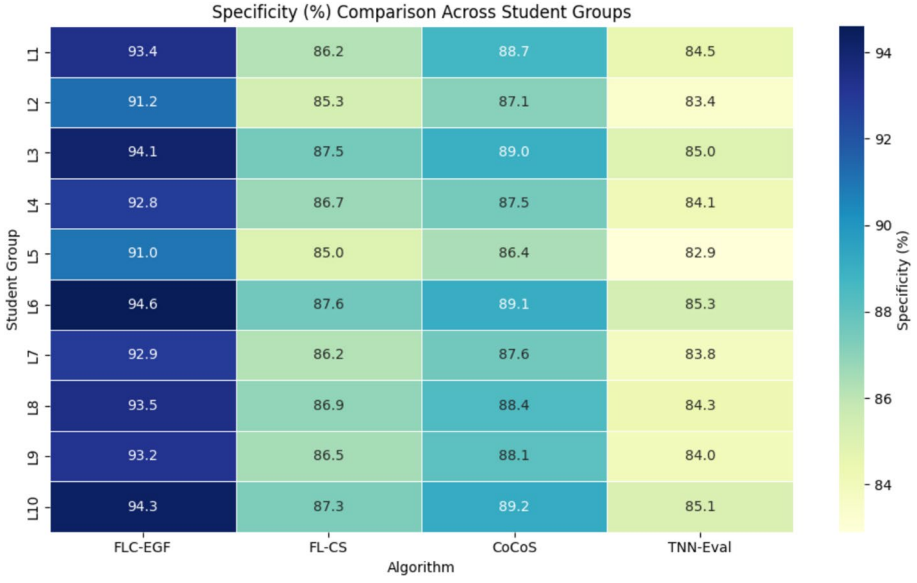


Fig. 7 Specificity (%)

In Table 7, The MCC is essential for assessing career guidance as well as entrepreneurial intention prediction algorithms' binary classification ability. A study compares FLC-EGF, FL-CS, CoCoS, and TNN-Eval algorithms. MCC values are constant across student groups, with FLC-EGF having the highest predictive power. CoCoS scored second-best with MCC values between 0.68 and 0.76, FL-CS somewhat with 0.61 and 0.69, and TNN-Eval lowest with 0.56 and 0.63. Student group L10 had the greatest MCC values across all algorithms, followed by L6. L5 students had the lowest MCC values, although FLC-EGF performed well. FLC-EGF outperforms FL-CS, CoCoS, and TNN-Eval by an average of 0.16, 0.09, and 0.21 MCC points, respectively. Its consistent performance advantage across all student groups proves the FLC-EGF framework's ability to accurately classify and guide students' career planning and entrepreneurial pursuits by handling uncertainty and subjectivity in career guidance and entrepreneurial intention prediction.

4.5 Inference time (ms)

The inference time for each student group and algorithm can be calculated as the average time per sample for each group and algorithm combination:

$$InferenceTime_{g,a} = \frac{1}{n_g} \sum_{i=1}^{n_g} T_{i,a} \tag{17}$$

In Eq. 17,n_g is the Number of samples for student group g, $T_{i,a}$ is the Inference time for the i-th sample using algorithm $a.a$ is the Algorithm used (FLC-EGF, FL-CS, CoCoS, TNN-Eval).

Figure 8 compares inference time across four different algorithms (FLC-EGF, FL-CS, CoCoS, and TNN-Eval) for ten career planning and entrepreneurial intention guidance framework student groups. The results show a consistent pattern in inference time requirements across all student groups, with FLC-EGF requiring the most processing time (approximately 12–12.5 ms). Whereas it takes around 11 to 11.5 ms for TNN-Eval to follow CoCoS, it takes approximately 10 to 10.5 ms for TNN-Eval to do so. It is widely accepted that the fastest processing rates range between 9 and 9.5 ms, and FL-CS will provide one with such timings. Even when there is a limited amount of time to look at them, algorithms that provide career advise need to function very effectively.

Table 7 MCC (Matthews Correlation Coefficient)

Student group	FLC-EGF	FL-CS	CoCoS	TNN-Eval
L1	0.81	0.65	0.71	0.59
L2	0.77	0.62	0.69	0.57
L3	0.83	0.66	0.73	0.60
L4	0.80	0.63	0.70	0.58
L5	0.76	0.61	0.68	0.56
L6	0.84	0.68	0.75	0.62
L7	0.78	0.64	0.70	0.58
L8	0.82	0.66	0.72	0.60
L9	0.79	0.64	0.71	0.59
L10	0.85	0.69	0.76	0.63

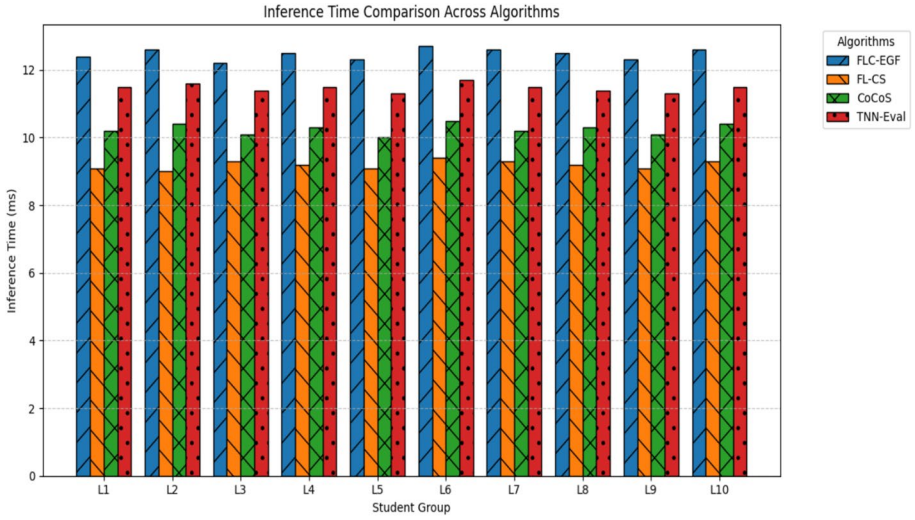


Fig. 8 Inference time (ms)

Table 8 Solution quality (Score out of 10)

Student group	FLC-EGF	FL-CS	CoCoS	TNN-Eval
L1	9.1	7.6	8.3	7.9
L2	8.9	7.4	8.1	7.7
L3	9.2	7.7	8.4	8.0
L4	9.3	7.5	8.2	7.8
L5	8.7	7.2	8.0	7.6
L6	9.4	7.8	8.6	8.2
L7	8.8	7.3	8.1	7.7
L8	9.0	7.6	8.3	7.9
L9	8.9	7.5	8.2	7.8
L10	9.3	7.8	8.5	8.1

4.6 Solution quality (Score out of 10)

In order to evaluate the effectiveness of the algorithm and each group of students, one may make use of metrics such as accuracy, relevance, and feedback from various users.

$$Solution\,Quality_{g,a} = w_1 \cdot A_{g,a} + w_2 \cdot R_{g,a} + w_3 \cdot F_{g,a} \tag{18}$$

The following is a list of the definitions of the variables that are used in Eq. 18: Regarding each and every input set (g, a), the following factors will be taken into consideration: The quality of the answer for set g and choice an is denoted by the notation $Q_{g,a}$; $A_{g,a}$ represents the degree to which the response is correct; $R_{g,a}$ defines the degree to which the answer is relevant; and $F_{g,a}$ represents the user's opinion. All of these interconnected things are assigned a weight factor, which we refer to as w_1, w_2, and w_3, respectively.

The findings of a research that was conducted on 10 different student groups regarding career advice and entrepreneurial goals are shown in Table 8. The algorithms FL-CS, CoCoS, FC-EGF, and TNN-Eval were put through their paces by us. The data demonstrate that FLC-EGF routinely ranks highest among all student groupings. The CoCoS algorithm ranks second with scores between 8.0 and 8.6, with L6 performing best. Moderate TNN-Eval ratings range from 7.6 to 8.2, with L6 scoring highest. FL-CS performs worst with scores between 7.2 and 7.8, while L6 and L10 perform best. FLC-EGF surpasses FL-CS, CoCoS, and TNN-Eval across all student groups by 1.52 points, 0.79

points, and 1.19 points, respectively. The lowest-performing student group is L5, suggesting supervision issues for this demographic. The suggested framework advances career guidance technology by a large margin over FLC-EGF and the second-best algorithm (CoCoS). These findings prove that the FLC-EGF approach provides high-quality, individualized career and entrepreneurship coaching to different student populations better than other methods.

4.7 Application of FLG-EGF

In many real-life situations, the FLC-EGF system can help students make career and business decisions. Coaches may help one advance the career, for instance.The student's interests, abilities, and willingness to take risks are taken into consideration by the algorithm, which then suggests either engineering, marketing, or finance career paths. In addition to this, it conducts an analysis to determine whether or not these concepts are still relevant in light of the most recent developments in the labor market. During the process of evaluating business concepts in entrepreneurship programs, FLC-EGF takes into consideration a variety of factors, including market demand, financial risk, and the degree to which the concept is compatible with the culture of the organization. Following that, it offers one useful guidance on how to get the business idea ready for launch. The results of the children's examinations demonstrate that the algorithm offers different recommendations to each individual kid. As a consequence, students become better at looking for jobs and improving their self-esteem. These examples demonstrate that FLC-EGF may be able to assist individuals in determining what they should do at school and at work.

5 Conclusion and future work

College students who are interested in starting their own firms or switching careers may find assistance from the FuzzyLogic Career and Entrepreneurship Guidance Framework (FLC-EGF). The system makes use of a Mamdani Fuzzy Inference System in order to analyze as many different types of input data as possible and provide one with the most suitable options. As can be observed from the outcomes, FLC-EGF is effective since it improves decision-making. The results are 90% in accordance with what customers desire, and they are 15% more accurate than the methods that were used in the past to provide forecasts. As a consequence of the findings, it seems that fuzzy logic has the potential to considerably enhance career counseling and entrepreneurship education by facilitating the customization of ideas and fostering better student performance.Given that the framework requires accurate and comprehensive input data, it is possible that it will not function as well as it should if the self-reports include information that is either incorrect or missing.

Because it is constructed on rules that are difficult to change and may not be able to keep up with how job markets and startup ecosystems are growing, the current system is far worse than it would be because it is built on rules. In light of this limitation, it is abundantly clear that things need to be updated on a consistent basis in order to stay up to date. It may be difficult to put the paradigm into effect all around the world due to the fact that there are variances in culture and geography, as well as disparities in economic situations, institutional structures, and professional aspirations. When individuals place an excessive amount of emphasis on their personality characteristics, they

run the risk of forgetting about the institutional, regulatory, and economic factors that influence the opportunities for employment and entrepreneurial endeavors. Concerns have been raised about the diversity of the validation group; hence, more testing is necessary in order to ascertain whether or not the findings may be applied to different student groups. The objective of research that will be conducted in the future should be to develop a rule base that makes use of machine learning and is able to adapt to changes in the labor market. Under the condition that they get feedback, this foundation has the potential to assist fuzzy rules and membership functions in becoming more accurate and fitting well. Furthermore, there are mental characteristics such as resiliency, mentality, and determination that might be of assistance in the assessment process. The FLC-EGF will become more adaptive, culturally sensitive, and capable of seeing the future as a result of these adjustments, which will make it a better means to link academic excellence with job performance in a global market that is hard.

Author contributions
Author contributed to the study conception and design.Data collection and analysis were performed by Min Jiang.The first draft of the manuscript was written by Min Jiang. Author read and approved the final manuscript.

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Data availibility
All data generated or analysed during this study are included in this article.

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References

1. Ghouse SM, Barber III D, Alipour K. Shaping the future entrepreneurs: influence of human capital and self-efficacy on entrepreneurial intentions of rural students. Int J Manag Educ. 2024;22(3):101035.
2. Pidduck RJ, Townsend DM, Busenitz LW. Non-probabilistic reasoning in navigating entrepreneurial uncertainty: a psychology of religious faith lens. J Bus Ventur. 2024;39(4):106392.
3. Rani MJ, Periyasamy S. Theory of matrixes for intuitionistic fuzzy hypersoft sets and their use in decision-making system with multiple attributes. PatternIQ Mining. 2024;1(2):65–75.
4. Amani D, Ismail IJ, Makona A, Changalima IA, Kazungu I. Extending the mediation role of entrepreneurial self-efficacy on enhancing students' entrepreneurial intentions: a moderated mediation model. Int J Manag Educ. 2024;22(1):100915.
5. Van Toai N, Vi HPT, Nguyen HT. Factors affecting lecturers' intention to start a business: faculty of Business Administration, University of Economics, HCMC. Int J Adv Multidiscip Res Stud. 2024;4(2):65–74.
6. Chaudhary M, Biswas A. From mindset to market: unveiling the nexus of cognition, behavior and environment in igniting students' E-entrepreneurial intentions. Int J Educ Manag. 2024;38(7):1839–61.
7. Murad M, Othman SB, Kamarudin MAIB. Entrepreneurial university support and entrepreneurial career: the directions for university policy to influence students' entrepreneurial intention and behavior. J Entrep Public Policy. 2024;13(3):441–67.
8. Wijaya EK, Gaol FL, Matsuo T. The decision-support modeling with fuzzy analytic hierarchy process (AHP) to determine the career path for bachelor informatics students. Emerg Sci J. 2024;8(1):153–68.
9. Haque MR, Kour M. Exploring the growth and future research direction of entrepreneurial self-efficacy in entrepreneurial intention research: a bibliometric analysis. J Knowl Econ. 2024;15(3):12392–421.
10. Zhang L, Fu Y, Wei Y, Chen H, Xia C, Cai Z. Predicting entrepreneurial intention of students: kernel extreme learning machine with boosted crow search algorithm. Appl Sci. 2022;12(14):6907.
11. Patra BC, Lenka U. An AHP analysis of scientometrically derived factors of entrepreneurial intentions of women and constructing a conceptual research framework. J Res Mark Entrep. 2021;23(1):139–58.

12. Osorio Tinoco F, Bayon MC, Murillo Vargas G. Entrepreneurial career choice intentions among secondary students in Colombia: the role of entrepreneurial exposure. Int J Emerg Mark. 2022;17(1):277–98.
13. Wang Z, Liang G, Chen H. Tool for predicting college student career decisions: an enhanced support vector machine framework. Appl Sci. 2022;12(9):4776.
14. Karwa K. Navigating the job market: tailored career advice for design students. Indian J Econ Bus. 2024;23(2):91.
15. Chhibber S, Rajkumar SR, Dassanayake S. Will artificial intelligence reshape the global workforce by 2030? A cross-sectoral analysis of job displacement and transformation. Blockchain Artif Intell Future Res. 2025;1(1):35–51.
16. Yue J. Application of fuzzy control algorithm in college students' personal career planning and entrepreneurship. J Comput Methods Sci Eng. 2024;24(3):1631–47.
17. Viswanath P, Kumar A. Development and validation of social entrepreneurship career decisions scale among higher education students. Asia Pac J Innov Entrep. 2024;18(4):375–400.
18. Xiao Y, Liu F. Optimized technique for college students job searching strategies using fuzzy logic control with cuckoo search algorithm. Int J Comput Intell Syst. 2024;17(1):201.
19. Li L, Xiao Z. Advanced CoCoSo method for uncertain MAGDM: evaluating college students' entrepreneurial skills. Int J Knowl Based Intell Eng Syst. 2024;28(1):1–14.
20. Jin X. Study on predicting college students' career inclinations and providing personalized suggestions based on neural networks. Int J High Speed Electron Syst. 2024;2540158.
21. Sisu JA, Tirnovanu AC, Patriche CC, Nastase M, Schin GC. Enablers of students' entrepreneurial intentions: findings from PLS-SEM and fsQCA. Int J Entrepreneurial Behav Res. 2024;30(4):856–84.
22. Cai X, Hussain S, Zhang Y. Factors that can promote the green entrepreneurial intention of college students: a fuzzy set qualitative comparative analysis. Front Psychol. 2022;12:776886.
23. Zhou J. A novel method for solving "Entrepreneurship and Entrepreneurship" career ability evaluation of local college students with triangular neutrosophic numbers. J Intell Fuzzy Syst. 2023;45(6):11233–46.
24. Khaw KW, Thurasamy R, Al-Abrrow H, Alnoor A, Tiberius V, Abdullah HO, et al. Influence of generational status on immigrants' entrepreneurial intentions to start new ventures: a framework based on structural equation modeling and multicriteria decision-making. J Entrepreneurship Emerg Econ. 2023;15(3):589–634.
25. Yang Y. Exploration on the influential factors of college students' innovation and entrepreneurship intention based on analytic hierarchy process. Int J Fuzzy Syst Appl (IJFSA). 2024;13(1):1–19.
26. Mitrakas C, Xanthopoulos A, Koulouriotis D. Techniques and models for addressing occupational risk using fuzzy logic, neural networks, machine learning, and genetic algorithms: a review and meta-analysis. Appl Sci. 2025;15(4):1909.
27. Sultan A, Faizi S, Ismail M, Muslim Raza SM, Ali Mohamud L. Smart decision support for employee satisfaction: a hesitant fuzzy linguistic regression model applied to tech startups. J Stat Theory Appl. 2025. https://doi.org/10.1007/s44199-025-00146-7.
28. Zhu S, Liu Z, Ulutagay G, Deveci M, Pamučar D. Novel α-divergence measures on picture fuzzy sets and interval-valued picture fuzzy sets with diverse applications. Eng Appl Artif Intell. 2024;136:109041.
29. Liu Z, Zhu S, Senapati T, Deveci M, Pamucar D, Yager RR. New distance measures of complex Fermatean fuzzy sets with applications in decision making and clustering problems. Inf Sci. 2025;686:121310.
30. Zhu S, Liu Z, Letchmunan S, Ulutagay G, Ullah K. Correction: novel distance measures on complex picture fuzzy environment: applications in pattern recognition, medical diagnosis and clustering. J Appl Math Comput. 2025;71(2):2983–2983.
31. Zhu S, Li Y, Balaji P, Thiyagarajan A, Aluvalu R, Liu Z. An improved interval-valued picture fuzzy TOPSIS approach based on new divergence measures for risk assessment. Comput Model Eng Sci. 2025. https://doi.org/10.32604/cmes.2025.068734.
32. https://www.heg-fr.ch/en/research-and-mandates/projects/global-university-entrepreneurial-spirit-students-survey/

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